

AFL LangGraph AI Agent — Executive Report

Domain-locked AFL chat + prediction assistant, evaluated and wrapped for deployment

Product Goal

Give AFL fans, editors, and stakeholders one conversational interface that answers exact historical questions (scores, player stats, head-to-head records), makes leakage-safe match/player/season predictions from trained models, explains AFL rules, and firmly refuses anything outside AFL — including attempts to override its own instructions — while carrying conversational context correctly across multi-turn sessions.

Architecture

A LangGraph state machine routes every message through: **load_context** → **router** → {**retrieval** | **prediction_match** | **prediction_premiership** | **prediction_player** | **direct_afl** | **refusal** | **clarification**} → **validation** → **response_formatter**. The router runs a deterministic keyword/regex classifier by default (100% accuracy, no external dependency) with an optional Gemini structured-output path when `GOOGLE_API_KEY` is configured, falling back automatically and silently on any LLM failure. Five retrieval tools and three prediction tools (match winner, player stat regressors for disposals/goals/kicks/marks/handballs/tackles, and a season-wide power-ranking built from the same match model) sit behind a numeric grounding check that verifies every figure in the final response actually traces back to the tool's own output. The whole app is wrapped in a FastAPI service (`POST /chat`, `GET /health`, `GET /router-status`) with structured JSON-line logging of every request.

Evaluation Results

Router accuracy: 100% (34/34 queries across every supported intent). **Prompt-injection scope guard:** 100% block rate (6/6 distinct injection styles tested, all held).

Category	Passed	Total	Pass Rate
conversational_coherence	5	5	100%
factual_qa	11	11	100%
prediction_sanity	4	4	100%
scope_guardrails	8	8	100%

28/28 combined evaluation cases pass across factual Q&A, prediction sanity, scope guardrails, and multi-turn conversational coherence.

Model performance vs. naive benchmark

Match-winner model (logistic_regression): test ROC-AUC **0.7143**. Against naive public benchmarks on the same held-out seasons [2024, 2025]: home-team-always-wins scores **56.8%** accuracy; a ladder-position-style proxy (better career win-rate entering the match) scores **57.9%**; the trained model scores **65.9%**. This is a real, if modest, edge over guessing — an honest picture of 'good enough' given AFL's large irreducible upset rate.

Player expected-disposals regressor (HistGradientBoosting): test MAE 3.89 disposals. Five additional stat regressors (goals, kicks, marks, handballs, tackles) are trained the same way.

Known Limitations

- Data recency: the most recent season in the data is 2025; premiership predictions for seasons beyond that are explicitly flagged as low-confidence extrapolations, not real forecasts.
- Model accuracy ceiling: match-winner ROC-AUC ~0.71 / test accuracy ~66% reflects AFL's genuine unpredictability, not a modeling shortcut — treat probabilities as directional signal, not certainty.
- No fixture/schedule file exists in the data, so single-match predictions can never confirm two teams are actually scheduled to play — this is disclosed in every such response.

- Guardrail edge cases: the scope guard is marker-based (broad by design to minimize false negatives); novel injection phrasings not resembling any known marker could still slip through and should be monitored via the weekly re-evaluation loop (see monitoring checklist).
- "Eligible roster" for player predictions approximates to players who appeared for a team in the most recent season present in the data — not a live, current team list.

Recommended Next Steps

- Add Elo-style dynamic team ratings and interaction features (recent form $\times$ rest days) to lift match-model ROC-AUC beyond ~ 0.71 , and run a full probability-calibration/reliability check.
- Stand up the weekly retraining loop described in the monitoring checklist as new rounds complete.
- Tighten FastAPI CORS origins and add authentication before any real external deployment.
- Expand the prompt-injection marker list based on real traffic (Task 4's off-topic/injection leak rate metric) rather than only the fixed test set used here.